

# Product Price History Tracker

## Multi-Platform Price Monitoring System

A backend-focused system to track **daily product prices across multiple e-commerce platforms** (Amazon, Flipkart, etc.) and expose them via REST APIs for frontend consumption.

Think **PriceBefore / Keepa**, but simpler, cleaner, and academically solid.

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## Project Objective

Track and store **historical prices** of products sold on different platforms so users can:

- See today's price
  - Compare prices across platforms
  - Analyze price trends over time
  - Make better buying decisions
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## Problem Statement

- Same product → different prices on different platforms
- Prices fluctuate daily
- Existing platforms show only current price, not structured historical data

This project solves that by maintaining a **normalized price history database** and exposing it via APIs.

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## Target Users

- End users looking for the best time to buy a product
- Developers consuming price-history APIs

*(Alerts, watchlists, and automation are future scope)*

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## System Architecture

```
Client (React)
  ↓ REST API
Spring Boot Backend
```

↓ JPA/Hibernate  
MySQL Database

- Frontend never interacts with DB directly
- Backend handles all business logic



## Database Schema



### Users

```
users
-----
id (PK)
name
email
password
```



### Products

```
products
-----
id (PK)
name
brand
category
image_url
```



### Platforms

```
platforms
-----
id (PK)
name
logo_url
```

## Product Price History (Core Table)

```
product_price_history
-----
id (PK)
product_id (FK → products)
platform_id (FK → platforms)
price
date
```

✓ One row = price of one product on one platform on one date

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## Example Data

**Product:** iPhone 15

| Date       | Platform | Price  |
|------------|----------|--------|
| 2026-01-10 | Amazon   | 72,999 |
| 2026-01-10 | Flipkart | 71,999 |
| 2026-01-11 | Amazon   | 71,499 |
| 2026-01-11 | Flipkart | 70,999 |

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## REST API Endpoints

### Get All Products

```
GET /api/products
```

---

### Get Product Details

```
GET /api/products/{productId}
```

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## Get Product Price History

```
GET /api/products/{productId}/price-history
```

### Sample Response:

```
[
  {
    "platform": "Amazon",
    "date": "2026-01-10",
    "price": 72999
  },
  {
    "platform": "Amazon",
    "date": "2026-01-11",
    "price": 71499
  }
]
```

## Get Today's Prices (Optional)

```
GET /api/products/{productId}/today-prices
```

## Frontend Responsibilities

Handled **entirely by frontend developer**:

- Product listing page
- Product detail page
- Platform-wise price comparison table
- Price history charts (Line Graph)
- API integration & error handling

 Backend responsibilities **do NOT include UI or charts**

## Tech Stack

### Backend

- Java
- Spring Boot
- Spring Data JPA
- MySQL

### Frontend

- React
- Axios / Fetch API
- Chart.js or Recharts

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## Suggested Project Structure

```
product-price-tracker/  
├── backend/  
│   ├── src/main/java/com/example/pricetracker/  
│   │   ├── controller/  
│   │   ├── service/  
│   │   ├── repository/  
│   │   ├── model/  
│   │   └── dto/  
│   ├── src/main/resources/  
│   │   ├── application.yml  
│   │   └── schema.sql  
│   └── pom.xml  
├── frontend/  
│   ├── src/  
│   ├── public/  
│   └── package.json  
├── database/  
│   └── schema.sql  
├── README.md  
└── .gitignore
```

## Project Status

-  System design completed
  -  Database schema finalized
  -  Backend APIs under development
  -  Frontend not started yet
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## Future Enhancements

- Price drop alerts
  - User watchlist
  - Scheduled cron jobs for daily price updates
  - Admin dashboard
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## License

This project is for **educational and academic purposes**.